

PRATHAM MITTAL

AI/ML Engineer · Full Stack Developer · B.Tech CSE, Expected May 2027

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PROFESSIONAL SUMMARY

Final-year B.Tech CSE student (CGPA 9.00) with hands-on experience building machine learning pipelines, full-stack applications, and CI/CD automation. Engineered a 3-model deep learning ensemble achieving 96.8% accuracy, designed 15+ REST API endpoints for a production open-source platform, and automated deployment pipelines reducing release cycles by 45%.

EDUCATION

VIT Bhopal University — Bachelor of Technology, Computer Science & Engineering | Expected May 2027 2023 – 2027

CGPA: 9.00 / 10.00 Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, Software Engineering, Machine Learning

TECHNICAL SKILLS

Languages	Python, Java, C++, JavaScript, TypeScript, SQL, C, HTML5, CSS3
AI/ML	TensorFlow, Keras, Scikit-learn, OpenCV, CNNs, Transfer Learning, Ensemble Methods, Model Validation
Frameworks	React.js, Node.js, Express.js, REST APIs, Spring Boot, Streamlit
Databases	MongoDB, MySQL, Redis, SQLite, Database Design, Query Optimization, NoSQL
AI Tools & Testing	GitHub Copilot, ChatGPT, Claude, Unit Testing, Code Review, Agile, SDLC

PROFESSIONAL EXPERIENCE

FOSSEE, IIT Bombay – Software Engineering Intern (Open Source) | On-site Feb 2026 – Present

- Designed and developed **15+ REST API endpoints** using Node.js and MongoDB, transforming technical specifications into operational building blocks for the OpenHW Studio simulation platform (openhwhw-studio.fossee.in).
- Diagnosed and fixed **30+ emulator simulation bugs** through systematic **code forensics** and root-cause analysis, improving system reliability across 25+ hardware component types.
- Reviewed and audited **20+ pull requests** in a 5-engineer team, actively hunting for **logical errors and edge-case failures** to enforce code quality standards via Git/GitHub.
- Collaborated with senior engineers to translate user requirements into scalable microservice architectures, following **test-driven validation** practices before merging features.

IBM – DevOps & Software Engineering Intern | Remote May 2025 – Jul 2025

- Architected **CI/CD pipelines** using Jenkins and GitHub Actions, automating the end-to-end **SDLC from build to deployment** and cutting release cycles by **45%**.
- Managed **50+ containerized deployments** with Docker and Kubernetes across **3 environments**, implementing **observability** tooling to identify performance bottlenecks.
- Provisioned infrastructure using Ansible, reducing environment setup time by **60%** and improving **production stability and system efficiency** for an 8-member cross-functional squad.

PROJECTS

Eye Disease Prediction – Machine Learning Pipeline | [GitHub](#) Sep 2025 – Dec 2025

Python, TensorFlow/Keras, OpenCV, Streamlit, Transfer Learning, Ensemble Methods

- Engineered a **3-model ensemble ML pipeline** (ResNet50, MobileNetV3, EfficientNetB0) with a majority-voting and confidence-scoring conflict-resolution algorithm to classify 4 retinal conditions from OCT scans.
- Implemented **test-driven model validation** with systematic accuracy benchmarking, confusion matrix analysis, and automated diagnostic PDF reporting using fpdf.
- Achieved **96.8% validation accuracy** across **2,000+ lines of code** and 20+ files, delivering a production-ready clinical diagnostic tool with a Streamlit dashboard interface.

Arduino Simulator – Full-Stack Web Application | [GitHub](#) Feb 2026 – Present

Node.js, React.js, MongoDB, REST API, Arduino C++, Asynchronous Logic

- Architected backend services and emulator logic for a browser-based simulator supporting real-time **parallel execution** across **25+ electronic component types** with asynchronous state management.
- Built end-to-end using Node.js and React.js, implementing component validation, state persistence, and automatic **C++ code generation** through object-oriented design patterns.
- Reduced emulator crash rate by **40%** by incorporating new debugging methods and **continuous optimization** practices within a 4-member collaborative team.

CERTIFICATIONS

Machine Learning | NPTEL, IIT Madras **The Full Stack** | Meta (Coursera) **DevOps Fundamentals** | IBM **Python** | HackerRank

ACHIEVEMENTS

- Top 10 Finalist** — NISM Ideathon 2025, National Institute of Securities Markets, India (national-level competition)
- Selected for **FOSSEE Open Hardware Internship, IIT Bombay (2026)** based on competitive project submission evaluated by senior engineers